Toxicology and risk assessment of coumarin: focus on human data.

Coumarin is a secondary phytochemical with hepatotoxic and carcinogenic properties. For the carcinogenic effect, a genotoxic mechanism was considered possible, but was discounted by the European Food Safety Authority in 2004 based on new evidence. This allowed the derivation of a tolerable daily intake (TDI) for the first time, and a value of 0.1 mg/kg body weight was arrived at based on animal hepatotoxicity data. However, clinical data on hepatotoxicity from patients treated with coumarin as medicinal drug is also available. This data revealed a subgroup of the human population being more susceptible for the hepatotoxic effect than the animal species investigated. The cause of the high susceptibility is currently unknown; possible mechanisms are discussed. Using the human data, a TDI of 0.1 mg/kg body weight was derived, confirming that of the European Food Safety Authority. Nutritional exposure may be considerably, and is mainly due to use of cassia cinnamon, which is a popular spice especially, used for cookies and sweet dishes. To estimate exposure to coumarin during the Christmas season in Germany, a telephone survey was performed with more than 1000 randomly selected persons. Heavy consumers of cassia cinnamon may reach a daily coumarin intake corresponding to the TDI.